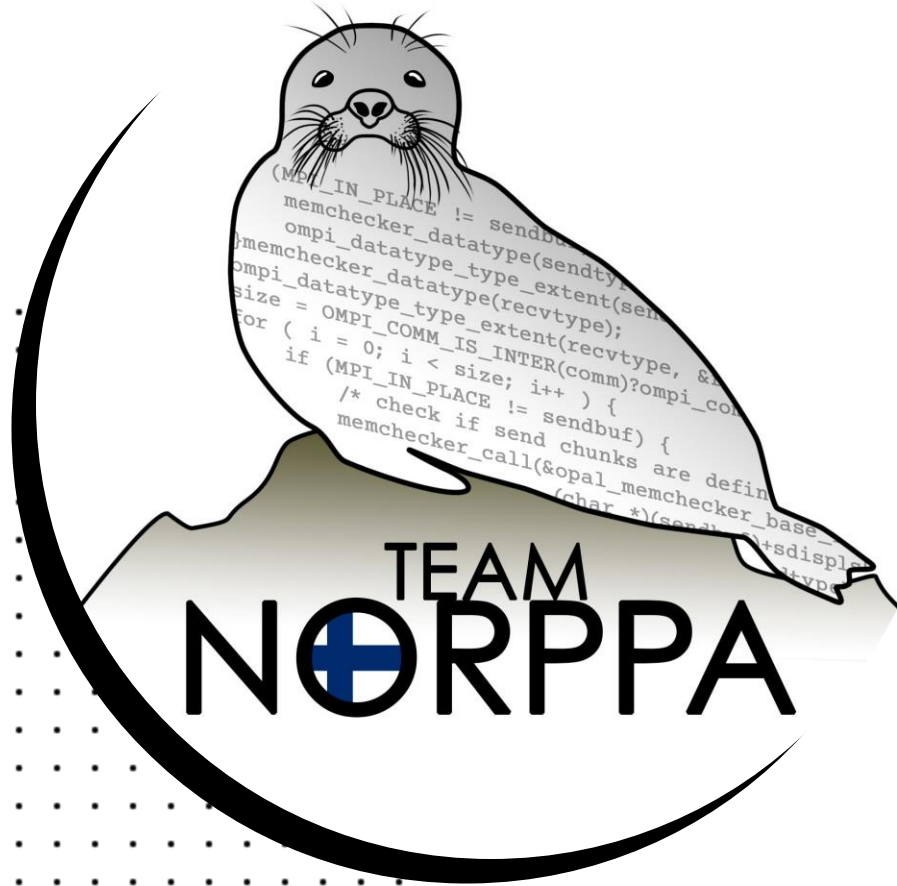


WELCOME TO THE FINNISH SCC TEAM!



Team Norppa's logo (2023)

- Today's agenda:
 - ISC Student Cluster competition in detail
 - Getting to know each other
 - Coffee break and snacks!
 - CSC's Supercomputing environment
 - Logging into Mahti

FIRST AND FOREMOST

- The advisors
- Contact information

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CREATING A USER THROUGH MYCSC

- Go to our GitHub page at <https://github.com/cschpc/student-cluster-competition-2024>
- Go to the folder: Training/01-kickoff
- Read the instructions and create a user through MyCSC
 - We'll use this to give you access to Mahti, Puhti and Lumi



SCC@ISC25

COMPETITION DETAILS





AS A REMINDER

- The aim of the competition
- On-site vs. Virtual
- Teams
 - Students
 - The role of the advisor
- ISC High performance
 - May 2025

ISC SCC TASKS
AND HARDWARE

WHAT DOES THE STUDENT CLUSTER COMPETITION AT ISC ENTAIL?

- Cloud resources or BYOC (virtual vs. on-site)
- Final competition
 - Booth and cluster setup
 - Benchmarking section
 - Application benchmarks
 - Mystery application section
- Webinar introductions to the competition applications
- Award ceremony and team interviews

ON-SITE VS VIRTUAL

Virtual

- Coding challenge
- Three scientific applications(2024)
 - Climate code
 - DFT simulations
 - Spectral element flow
- Using a remote supercomputer

On-site

- Same scientific applications
- + 3 Different benchmarks
 - HPL
 - HPCG
 - HPC Challenge
- + Secret application
- Each team brings their own cluster

THE CLOUD SYSTEM (VIRTUAL COMPETITION)

- DKRZ and Bridges-2 Supercomputers in '24
- CPU and GPU partitions
- Some applications on one system and some on the other
- Accessed and used similarly to CSC's supercomputers



ON-SITE SYSTEMS

- Consists usually of a few high-end servers
- Maximum power draw 6kW (2024)
- 2-4 servers create a mini-cluster
 - Connected with fast interconnect
- Assembled into a server rack a day before benchmarking the system



BENCHMARKING

- Only on-site
- Testing the system before the competition
- Three benchmarks in total
 - HPC Challenge (HPCC)
 - High-Performance Linpack (HPL)
 - High-performance Conjugate Gradients (HPCG)
- These will be prepared for in advance



THE 48-HOUR FINAL COMPETITION



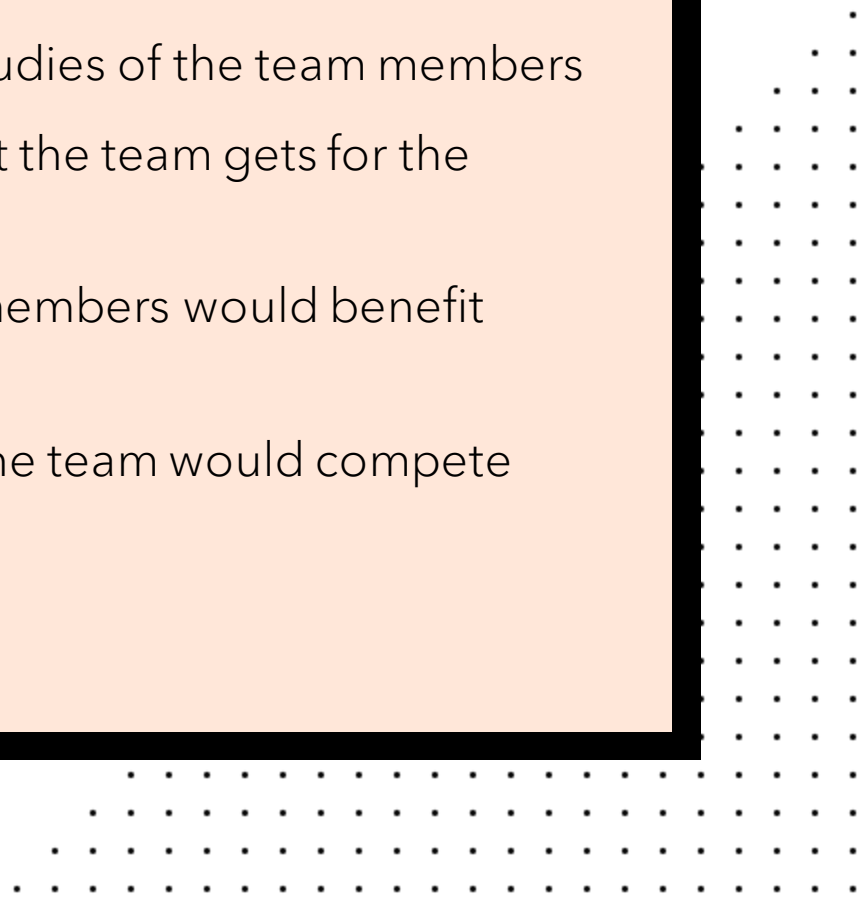
- Over 3 days between 8am-11:59pm on conference days
 - Non-stop in the virtual version
- Applications are run in separate timeslots in '24



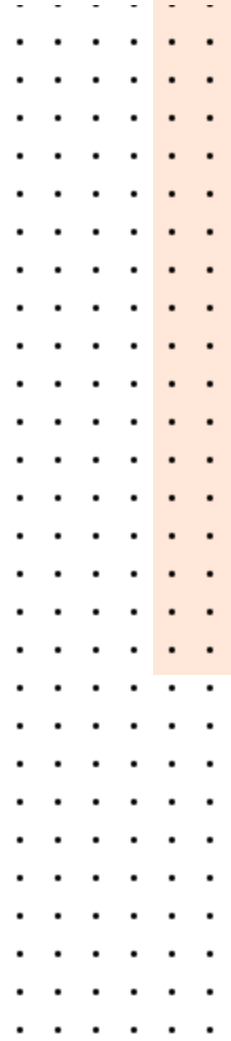


APPLICATION PERIOD: AUGUST 2024

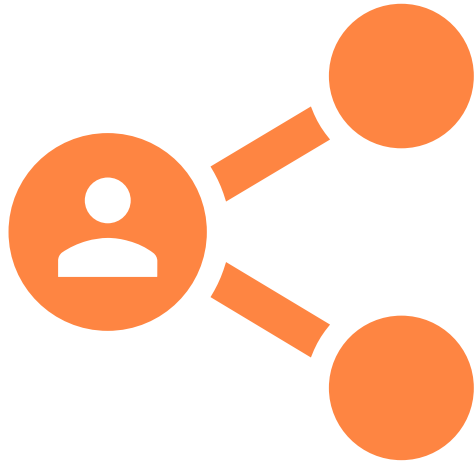
In the application there are sections asking about:

- Team members
 - The skills and studies of the team members
 - The support that the team gets for the competition
 - How the team members would benefit from attending
 - Hardware that the team would compete with
- 

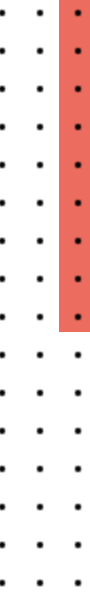
Q U E S T I O N S
A B O U T T H E
C O M P E T I T I O N
D E T A I L S ?



GETTING TO KNOW EACH OTHER



- Task: In pairs, give us a short introduction of the other person
- Be sure to ask some of the following:
 - Their name (important!)
 - Their study track
 - Their hobbies or interests
 - What do they hope to learn from the student cluster competition?



LET'S DISCUSS PRACTICALITIES

- Preferred platforms for communication?
 - Telegram
- Meeting schedule and preferred days
 - Next meeting 25.3.

1 5 M I N
B R E A K !

